Automated Estimation of Functional Walking Test Performance Using Video-Based Pose Estimation

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1 Intro

1.1 Why The Ten Meter and Two Minute Walk Tests

Functional status pertains to an individual's ability to perform daily living activities that enable "physical, psychological, social, spiritual, intellectual" roles (Wang, 2004). Walking speed is routinely used to assess functional status as well as the overall health for a range of different populations (Middleton, 2013). Among the most widely used field based assessments are the 10 Meter Walk Test (10MWT) and 2 Minute Walk Tests (2MWT), which provide clinically meaningful measures of walking speed and endurance. The 10MWT and 2MWT are benchmark field tests for evaluating functional mobility in clinical and rehabilitation contexts. Although their clinical utility is established, they are widely used to evaluate the functional status, progress, and possible fall risks in populations with impairments (Lindholm, 2018, Middleton, 2013). These assessments are traditionally administered in person and rely on manual timing and observation. Additionally, this limits the frequency of tests that can be conducted due to clinician time and cost, as well as patient burden. In person testing can be burdensome for patients with mobility limitations and may restrict access for individuals in rural or underserved areas.

1.2 Advantages of Remote Video Assessment

There are also widespread initiatives within the National Institutes of Health devoted to digitizing and making clinical trials more pragmatic and reflective of real-world conditions [61-63].

Many of the same barriers (e.g., transportation, cost, burden) to clinical assessments are also the drivers of low participation in clinical trials (e.g., only 8% of cancer patients enroll in trials) and individuals who are not near a site find participation cost prohibitive, exacerbating disparities in trial participation [61,64]. Optimal assessment of outcomes in clinical trials may ultimately include a combination of remote and in-person assessment, which may expand recruitment, reduce patient burden, and facilitate retention [57].

Recent advances in computer vision and the widespread availability of low cost video have created new opportunities for these functional assessments to be remote. However, achieving high accuracy with remote assessments still poses a challenge. Automating these tests via video based analysis could allow clinicians to remotely and objectively evaluate patients' functional mobility.

The present project, PathML, seeks to address these gaps by leveraging computer vision methods, primarily through pose estimation, to automate the scoring of standardized functional assessments. By automating the scoring of standardized functional tests such as the 10MWT and 2MWT, PathML aims to enable accurate remote assessment of functional mobility. In this study, we present methods for estimating walking velocity and total walking distance from video across multiple datasets and clinical contexts. This supports our broader goal of bringing video based functional assessment into telehealth and clinical research workflows.

1.3 Other Work

Recent work has explored video based methods for estimating velocity and distance covered in functional walking tests such as the 10MWT. Boripuntikal et al. (2024) argues against the traditional motion capture methods and states that they don't capture gait speed at specific times, rather they just get the average velocity of the total test. The test set up was exactly the same as the PathML set up being that there were four points being the start, 2 meters, 8 meters, and finish, for the sake of using the middle 8 meters to analyze gait speed. The video was split up by frame, the body of each person was detected by using a “background subtract technique.” Here there was a subtraction value per pixel, the “large subtraction value” was used to define the foreground, and the “small subtraction value” was used to define the background. Using these values a two-dimensional median filter was used to reduce the noise. Then the “BLOB Analysis Bounding Box” was employed to pinpoint the location of the human body, where a centroid was determined to calculate the center of mass in both horizontal and vertical directions using a “two point lamina.” Boripuntikal et al. compared their computer vision based system against the traditional motion analysis system by comparing the results of the computer vision based system with the motion analysis system. This test was repeated 30 minutes later in order to evaluate the test-retest reliability. The average gait speed from both the computer vision base system and the motion analysis system showed high to very high agreement with correlation ranging from “0.7 to 1.00 (P-value < 0.05).” The computer vision system also held a very high repeatability with “ICCs falling between 0.94 and 0.986,” with a total SUS score for all participants being 75.83. These results demonstrate that a computer vision based system can get clinically meaningful accuracy. The approaches used in Boripuntikal et al. (2024) differ from our own. When approaching the smoothing of the keypoint jitter, the foreground - background two dimensional median filter may be something that we could try in order to increase our own accuracy.

(ACM) aimed at a gait speed detections system for finding instantaneous walking speed of older adults. A standard camera fixed on a tripod with 3-way head was used to collect the body motion from the start and end points within the 10 meter segment, this being 6m apart.

Frames were extracted, the background was subtracted, and then the color was converted to greyscale and binary image. In order to reduce noise, a two dimensional median filter was employed. The centroid was calculated as well as the start and end points. Then speed was calculated using the distance traveled over time. There was good correlation between this approach and the motion analysis system with a p value < 0.0001. This again demonstrates that a computer based approach yields good results. However, this approach is not designed for real world remote deployment and it relies on the fact that the participants wear motion capture suits and have a completely stable camera.

Stenum et al. (2024) also successfully applies pose estimation based video analysis across populations, focusing on different diseased groups. This further demonstrates that pose tracking can accurately support gait analysis outside of traditional marker based systems. Though, this study focuses on step changes within diseased populations rather than the scoring of a functional test such as the 10MWT or 2MWT.

2 Methods

2.1 Data Collection Procedures

Data collection occurred at three primary sites: California Polytechnic State University (Cal Poly), University of Wisconsin–Milwaukee (UWM), and the James A. Haley Veterans Hospital (Tampa VA). Standardized annotation protocols and data dictionaries were developed collaboratively by Co Is Keadle, Strath, and Phillips to ensure consistent labeling across sites. Nine research assistants were trained and certified (>90% agreement required) to annotate test videos. In total, over 1,500 functional test videos were recorded and annotated.

Annotation focused on both categorical outcomes (e.g., pass/fail, task type) and continuous measures (e.g., time to completion, joint angles). These annotations provided ground truth for model development and evaluation and are described in detail in the Appendix - Table 2.

To assess annotation reliability, 25% of videos were dual-coded to examine inter-rater reliability across research assistants, which were excellent. We also conducted an analysis to compare the labeled outcomes (which were used for model training) to what the trained clinician/research assistant recorded at the time of the test and found they were excellent (ICC= 0.86-1.0), the video labels tended to be slightly faster. For example, TUG score was 0.6 seconds faster on video for Cal Poly and UWisc, which was attributed to the increased precision from using video where we can examine the exact frame the participant sits down compared to the clinical score that requires real-time determination of when the test ends and manually pressing a stopwatch. Collectively, these results show we successfully implemented an annotation protocol that provided consistent, accurate ground-truth labels for use in training and evaluation of the machine learning models.

2.2 Datasets

Three datasets were used in this analysis: Tampa VA (N=60 clinical participants; 818/818 functional tests annotated, 100%), Cal Poly (N=30 healthy participants; 408/408 tests annotated, 100%), and UWM (N=60 participants across functional statuses; 303/303 tests annotated, 100%).

The Cal Poly dataset consists of approximately 284 hours of annotated video collected under PI Keadle across multiple structured research protocols. Participants ranged in age and functional ability and completed standardized functional assessments including the 10 Meter Walk Test (10MWT) and 2 Minute Walk test (2MWT). Video recordings were captured using wearable GoPro Hero 5 cameras and later switched to Microsoft Surface Pro devices to test cross device transferability. In dataset three, dual camera recordings were collected for approximately 45 minutes per session using both devices, allowing for comparison between the two cameras.

For the Tampa VA data, 60 participants were recruited from the veteran and community populations. Participants were filmed using an iPad camera while completing the 10 Meter walk test (10mWT) and 2 minute walk test (2MWT). In total, 20 of the participants were considered healthy (no known functional limitations). In addition to completing the tasks, each health participant also completed two additional examples of common deficiencies or compensations for each test. An additional 20 veterans from outpatient amputee rehabilitation or cancer rehabilitation clinics were recruited to complete the study protocol at the VA hospital with a trained research assistant. An additional 20 patients were recruited from telehealth visits from outpatient amputee or cancer rehabilitation clinics. These visits took place remotely at the community-based outpatient centers (CBOCs). During telehealth visits, an attendant was present to assist with technical setup and patient safety. Patients were asked to perform simulated outcome assessments to compare to video assessments taken in outpatient clinics.

The University of Wisconsin dataset included 60 participants ranging in age from 21-100 years with varying levels of functional ability. The current study sample was stratified by age and gender, and we continuously evaluated recruitment across functional groups to maintain equal cell/cluster distribution. We drew from a multitude of ambulatory clinical patient groups including but not limited to the following: Cancer survivor, stroke, multiple sclerosis, Parkinson's disease, arthritis, chronic pain, low-functioning, general population. Populations were drawn and screened in local clinics and from institutes specific to functional/disease state, as well as through general advertisements and mailings. Participants were filmed (using Microsoft surface pro) completing a series of functional measures of upper/lower impairment that are used to stratify participants into one of three categories, normal, impaired or functionally limited. The specific tests are shown in Table 4 and described below.

For each participant, multiple trials of each test were recorded according to standardized protocols. A total of 139 10MWT trails were used from the VA dataset and 50 2MWT trails from CP and VA datasets. These datasets were previously annotated and served as training and validation data for our video based functional assessments. For the present Aim 2 analysis, only 10MWT and 2MWT videos were included.

2.3 Table:

Table 1. Participant Characteristics by Dataset

Site	N	Population Description
Cal Poly (Phase 1 / existing)	20 Healthy adults (18-85y)	Healthy adults completing standardized functional testing

UWM (new)	60 adults (18-100y)	20 healthy, 20 with some limitations, 20 functionally limited
Tampa VA (new)	60 adults	20 healthy, 40 amputee and/or cancer rehabilitation

2.4 Description of Computer Vision Methods

The 10MWT and 2MWT pipelines process walk trial videos in order to estimate the velocity or distance of the subject using pose keypoints extracted using MediaPipe Pose and we used the 3D world landmark output (pos_world_landmarks), expressed in meters. We used the Heavy variant of the MediaPipe model in order to restrict inference to a single person. Videos were processed at their own frame rate extracted from the file metadata and stored in a HDF5 file.

For the 10MWT keypoints are smoothed to reduce jitter and then evaluated against ground truth values for both 10m and the 2-8 m middle segment. Velocities are computed from the midpoint between the left shoulder key point (landmark 11) and right shoulder keypoint (landmark 12) and using the average of these via $p_t = \frac{(LS_t + RS_t)}{2}$. Coordinates are then smoothed using a Savitzky Golay filter with a window length of 15, polynomial order of 3, and then applied independently to x, y and z. Frame to frame displacements (m/frame) are converted to m/s by multiplying by the video's FPS and displacements are clipped to reasonable ranges $v_t = \left|\left|P_t + 1 - P_t\right|\right| \times FPS$, In order to further smooth, we clipped the velocity to stay within 0.3 and 1.25 m/s to remove spikes caused by pose jitter. For the 2-8m middle segment, we did not use the actual cones, rather a fixed fractional frame window of the start 5% and the end 95% of frames was used to approximate the segment, we then averaged the velocities in that region, added a constant offset of +0.45 m/s, and clipped the velocities between 0.4 and 3.0m/s. This constant was added due to the pattern of predicting slower middle 2-8m sections when compared to ground truth. Predictions are compared to annotated ground truth (VA) to compute absolute error $AE = \left|v_{pred} - v_{gt}\right|$, average percent $\%Error = \frac{\left|v_{pred} - v_{gt}\right|}{v_{gt}} \times 100$, and median percent errors for both 10m and 2-8m distances. Batch processing skips known poor trials, generates velocity-time plots, logs per-trial results to 10mwt_eval_log.txt, and overall accuracy is written as well.

For the 2mwt we took a completely different approach as multiplying the computed velocity by time was extremely inaccurate with an initial reading of 59.9% error. Due to the nature of the test, the camera angle held a much more frontal plane view, causing us to find a completely new method of calculating distance traveled. Instead of net displacement, total walking distance was found by summing directional changes in the keypoint trajectory to mimic the number of cone stretches achieved. Distances between turning cones were summed to reflect the total direction changes and remaining walk time for a final incomplete cone distance was projected using the average velocity from the previous cone passes.

$Total\ Distance = (N_{legs} \times D_{cone}) + D_{partial}$ where N_{legs} = number of completed cone to cone passes, D_{cone} = fixed distance between the cones, and $D_{partial}$ = distance covered during the final incomplete leg. $D_{partial} = v_{robust} \times remaining\ time$ and robust velocity is median/IQR filtered from previous legs. To detect turns, the distance between the left and right shoulder points $d_t = \left\|RS_t - LS_t\right\|$ was used to identify when the subject was walking towards and away from the camera. The peaks and valleys of these oscillations marked cone passes.

A Savitsky Golay filter with a window of 21 and polynomial order of 3 was also used to reduce jitter of the key points to ensure a more accurate read on the oscillations.

2.5 Evaluation Metrics

In order to evaluate the pipelines, we used optimisation metrics of mean and median percent error where Percent Error = $\frac{|Prediction - GroundTruth|}{GroundTruth} \times 100.$

3 Results

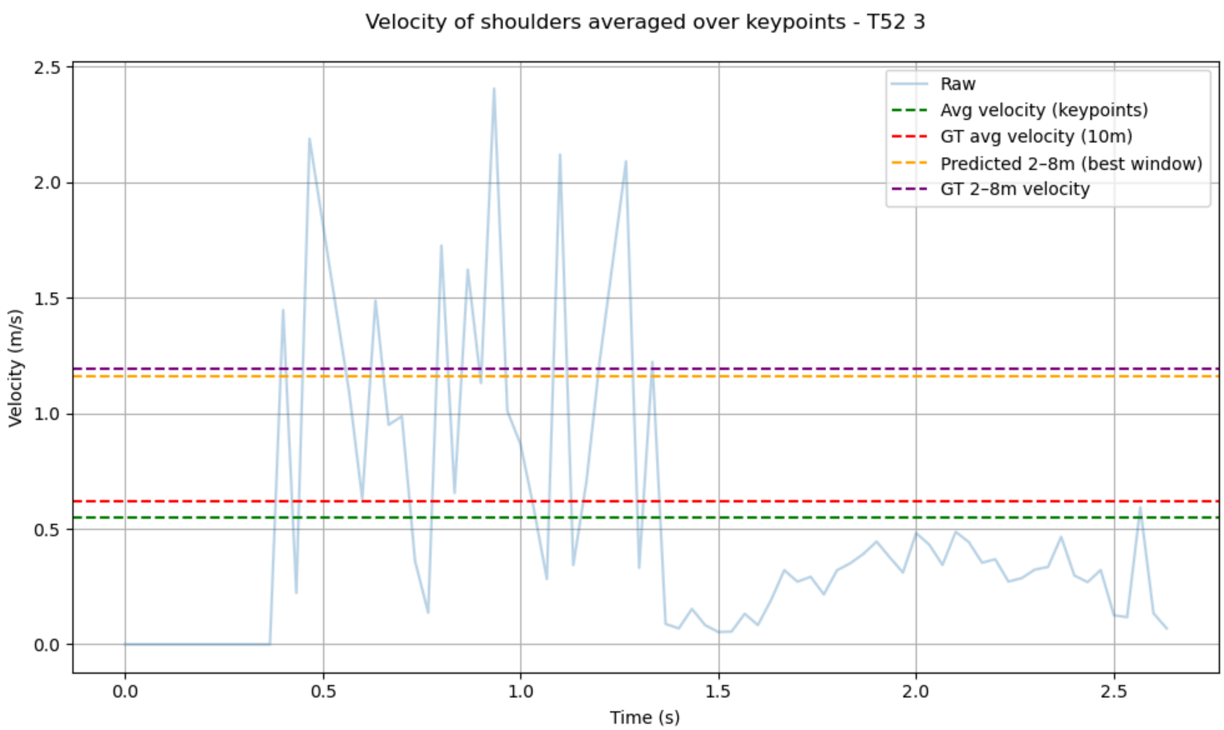
3.1 10MWT Results

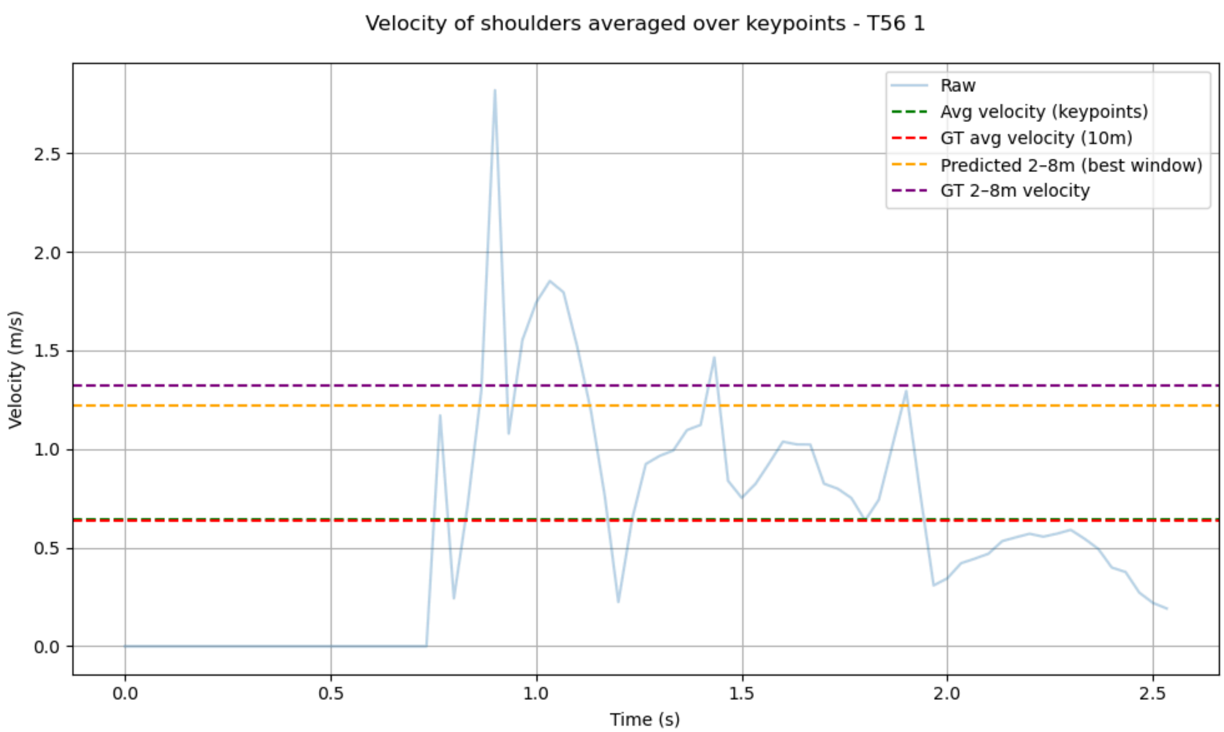
For the 10MWT we evaluated the average and median absolute error as well as percent error which was calculated by comparing the predicted distance traveled and the ground truth for a total of 139 trials from XX Datasets. Predicted velocities were compared to the annotated ground truth.

For the 10 m segment the model achieved an mean absolute error of 0.112 m/s, a median absolute error of 0.089 m/s, an average percent error of 19.0%, and a median percent error of 15.1%. For the 2-8 meter sections the model achieved an average absolute error of 0.233 m/s, a median absolute error of 0.198 m/s, average percent error of 22.0%, and a median percent error of 17.9%.

10MWT:

Metric	10 m Value	2-8 m Value
Avg Abs Error (m/s)	0.112	0.233
Median Abs Error (m/s)	0.089	0.198
Avg % Error	19.0%	22.0%
Median % Error	15.1%	17.9%





3.2 2MWT Results

For the 2MWT we evaluated the average and median absolute and percent error on a total of XX trials across the VA, Cal Poly, and UWisc datasets. Predicted velocities were again found by comparing our predictions against the annotated ground truth.

The model achieved an average absolute error of 95.643 ft, median absolute error of 83.0 ft, average percent error of 20.9%, and median percent error of 16.0%.

2MWT:

Metric	2 min Value
Avg Abs Error (ft)	95.643
Median Abs Error (ft)	83.0
Avg % Error	12.4%
Median % Error	16.7%

3.3 Model Improvement

The initial implementation of the 10MWT pipeline produced an average percent error of 60.6% and 82.8% median percent error for the full 10 meter segment. After refining smoothing as well as adding in velocity clipping thresholds, mean percent error decreased to 19%. Similar improvements were seen in the 2-8m segment (66.7 to 21.7).

For the 2WMT, the initial velocity based approach yielded a mean percent error of 59.9%. After implementing the oscillation based cone detection method, mean percent error dropped to 12.4%.

10MWT

	AVG % Error	Median % Error
Starting Results 10m	60.6%	82.8%
Current Results 10m	19.0%	15.1%
Starting Results 2-8m	66.7%	48.3%
Current Results 2-8m	21.7	18.0%

2MWT

	AVG % Error	Median % Error
Starting Results 2MWT	59.9%	59.5%
Current Results 2MWT	12.4%	16.7%

Algorithmic refinements substantially reduced prediction error across both tasks, with an average percent error decreasing from 60.6% to 19.0% for the 10MWT and from 59.9% to 12.4% for the 2MWT.

4. Discussion

4.1 Summary of main findings

The 10MWT was able to yield a result of 19% mean error, while the 2MWT yielded a performance of 12.4% mean error. The results improved significantly after adding in smoothing and clipping into the 10MWT pipeline, taking the error from 60.6% to 19%, as well as utilizing the oscillation technique in the 2MWT pipeline which brought the error from 59.9% down to 12.4%. Together, these results demonstrate that pose based analysis of standard video recordings of clinical walking tests can produce reasonably accurate estimates of gait speed and walking distance.

4.2 Interpretation of 10MWT findings

We focused on finding the 2-8m section velocity in order to account for acceleration and deceleration, as is common with the test. Smoothing greatly increased accuracy as the MediaPipe keypoints naturally had a lot of jitter, making their movements in space jump around unnaturally. Using the Savitzky Golay, we were able to greatly reduce this jitter and replicate a much more natural movement capture. In order to further reduce large unnatural spikes in velocity, introducing a clipping procedure took care of what smoothing could not. A key component of the 10MWT and 2MWT pipelines was the use of MediaPipe's 3D world landmark output, this is expressed in meters rather than image coordinates. The 2D image-space coordinates are influenced by camera position, zoom, and perspective distortion, whereas the 3D world landmarks use spatially scaled positions that remain more stable across frames. This reduced sensitivity to camera movements as well as subject distance from the camera, further reducing jitter.

There was also a 2-8m section bias towards underprediction, this is likely due to smoothing dampening peak velocities. We used a higher clipping window for the middle section in order to help counteract this.

The mean absolute error of 0.112 m/s for the full 10 meter segment is relevant in clinical contexts as in many rehabilitation populations, a change in ~0.1 m/s in gait speed is considered a minimally clinically important difference (MCID). Therefore, our error is close to the threshold of clinically meaningful change, suggesting the method may be useful in practice. While there are a lot of changes and improvements that could be made, these results suggest that video based estimation may be capable of detecting clinically significant differences in mobility when implemented correctly.

4.3 Summary of 2MWT Findings

Initially, the 2MWT pipeline utilized a keypoint displacement approach similar to the 10MWT pipeline, though this yielded very poor results likely due to the camera angle capturing the frontal plane. The camera primarily captures lateral displacement rather than forward displacement. In order to counteract this we shifted our approach to monitor the oscillations between the two shoulder keypoints. This better reflected the structure of the test as the model explicitly counted cone to cone transitions and accumulated distance accordingly. The alignment between the model structure and the clinical test design likely explains the substantial reduction in error observed following refinement.

4.4 Comparison to Prior Work

Prior work has demonstrated that computer vision based systems are reliable in capturing gait speed with a strong agreement to motion analysis systems. Boripuntikal et al. (2024) uses background subtraction, centroid tracking, and median filtering to estimate gait speed for the 10MWT and was able to report high correlation with motion capture systems ($r = 0.7-1.00$). Similarly, the ACM based system which was developed for older adults was able to report high correlation coefficients between 0.936 and 0.987 when compared to laboratory motion analysis. Sternum et al. further shows that pose estimation systems can accurately predict gait parameters outside of traditional marker based systems.

While these studies show promise, most were conducted in a controlled laboratory environment with stable cameras, and small sample sizes. To contrast this, the PathML study was done in a much more realistic environment, across 3 datasets, including healthy individuals and clinical populations. Recordings were captured across multiple devices and telehealth settings. Rather than strictly comparing against motion capture suits, this work focused on reflecting real world conditions.

Additionally, while the other studies utilized 2D image based centroids or bounding box tracking, this study leveraged 3D world landmarks from MediaPipe Pose, expressed in meters. This helps us ignore camera specific calibration procedures and allows for direct spatial scaling. Together, these approaches are more aligned with a scalable, remote clinical implementation.

4.5 Clinical and Telehealth Implications

Automating the 10MWT and 2MWT using video based pose estimation has important implications for remote clinical care. These tests are commonly used to indicate functional status, fall risk, and rehabilitation progress, though they require on site evaluation. For individuals with mobility issues or who live far from an on site location may find repeatedly traveling for assessments burdensome.

The present findings suggest that video based estimation of walking velocity and total walking distance can approach clinically meaningful accuracy, particularly in the 10MWT where mean absolute error was near the threshold of minimally clinically important difference. By reducing reliance on specialized equipment, automated video analysis may enhance scalability and consistency in functional mobility assessment.

4.6 Limitations

There are several limitations to be considered. First the 2-8 meter segment of the 10MWT was approximated using a fixed fractional frame window (5%-95%) rather than automated detection of the actual physical cones. While this approach simplified implementation, it assumes constant walking behavior across trials and may not generalize to irregular pacing.

Second, velocity clipping thresholds as well as the constant added to the 2-8m segment were empirically selected based on observed prediction bias. Although these refinements reduced error, they were dataset informed, so the adjustments may not generalize across new populations.

Finally, the 2MWT oscillation based approach assumes a consistent frontal camera orientation. Performance may degrade in videos with substantial obstruction of view or camera instability.

4.7 Future Directions

Future work should focus on improving generalizability and automation. Automated cone detection using object detection frameworks could replace the fixed fractional window of the 10MWT, allowing for a much more generalizable and procedure representative model.

Although the datasets included multiple different functional populations, performance was not tested on specific populations. Future work could evaluate subgroup performance to see if adjustments are required for certain populations.

Ultimately, integration of remote functional mobility scoring into telehealth platforms may allow clinicians to more easily monitor the progression of their patients. This may allow them to identify functional decline earlier, as well as reduce barriers for individuals needing care.

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